6.4 Solving the Quartic

WORKSHEET

The question

STEP III 2017, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s3-q3

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself How many ways are there to split four objects $\alpha, \beta, \gamma, \delta$ into two pairs?

STAGE 1 The missing coefficient

Covers: finding A

Ask yourself How many ways are there to split $\alpha, \beta, \gamma, \delta$ into two pairs?

Pause and try In $y^3 + Ay^2 + By + C = 0$, what is the sum of the roots? And what is the sum of the three quantities $\alpha\beta + \gamma\delta, \ldots$, in terms of the quartic's coefficients?

STAGE 2 The specific cubic

Covers: Part (i)

Ask yourself Substitute $p = 0$, $q = 3$, $r = -6$, $s = 10$. Try small divisors of the constant term.

STAGE 3 Pairs of roots

Covers: Part (ii)

Pause and try $(\alpha + \beta)(\gamma + \delta)$ is the sum of all six pair products minus one of them. Also, what is $\alpha\beta \cdot \gamma\delta$?

STAGE 4 Solving the quartic

Covers: Part (iii)

Pause and try You know the sum and product of $\alpha+\beta$ and $\gamma+\delta$. To decide which is which, write the sum of the triple products in terms of $\alpha\beta$, $\gamma\delta$, $\alpha+\beta$ and $\gamma+\delta$, and use r.

What's really going on

Why the trick works: permutations of roots

In 1770 Joseph-Louis Lagrange asked why the methods for cubics and quartics work, and found the answer in the *permutations* of the roots. There are 24 ways to rearrange four roots. Under all of them, the three quantities $y_1 = \alpha\beta+\gamma\delta$, $y_2 = \alpha\gamma+\beta\delta$, $y_3 = \alpha\delta+\beta\gamma$ are merely shuffled among themselves: each rearrangement of the four roots induces a rearrangement of the three pairings (Figure 6.4.1). So they satisfy a polynomial of degree 3 with coefficients symmetric in the roots. In modern language, there is a group homomorphism from the symmetric group S_4 (all 24 permutations) onto S_3 (the 6 permutations of the pairings), and that is why a quartic can be reduced to a cubic.

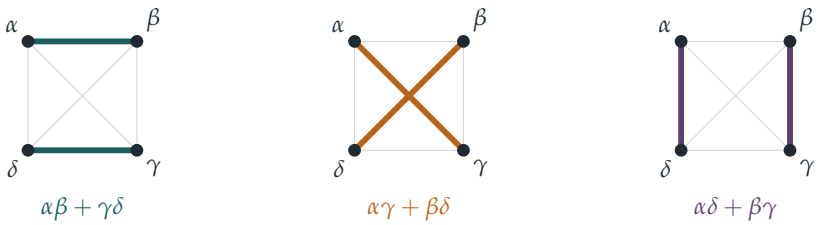


Figure 6.4.1 The three ways to split four roots into two pairs, and the corresponding roots of the resolvent cubic. Any rearrangement of $\alpha,\beta,\gamma,\delta$ permutes these three pictures among themselves, which is why the three quantities satisfy a cubic with coefficients expressible in terms of the quartic's.

The quintic

For five roots, Lagrange looked for a similar quantity taking only a few values under permutations of the roots, and could not find one that helped. In 1799 Paolo Ruffini, and in 1824 Niels Henrik Abel (completing the proof), showed that there is no general formula for the roots of a quintic using only arithmetic and radicals. A few years later Évariste Galois explained exactly which polynomials can be solved by radicals: those whose *group of symmetries of the roots* can be broken down in a particular way into simpler pieces. The symmetric group S_4 can be (via S_3 and smaller groups), but S_5 cannot, because it contains a "simple" piece of size 60 that cannot be broken down further. Galois wrote up his theory in a hurry the night before he was killed in a duel in 1832, at the age of twenty. It is now one of the jewels of a mathematics degree, usually met in the third year.

Solving cubics

The cubic itself can be solved by a similar trick: write a depressed cubic $x^3 - 3px + q$ as a combination of two perfect cubes, which reduces it to a quadratic. That is the method of the 2011 question in the exercises, and it is equivalent to Cardano's formula. When the cubic has three real roots, the formula still requires complex numbers along the way, the phenomenon that forced mathematicians to take complex numbers seriously (see the introduction to Chapter 5).

Try this next

1. **Check the cubic.** Verify that $y_1y_2 + y_2y_3 + y_3y_1 = pr - 4s$, where $y_1 = \alpha\beta + \gamma\delta$ etc. (Expand and use $\sum\alpha = -p$, $\sum\alpha\beta\gamma = -r$, $\alpha\beta\gamma\delta = s$.)

2. **The cubic as a sum of cubes.** Try STEP III 2011, Question 3 (step.maths.org/questions/11-s3-q3). It solves cubics by writing them as a combination of two cubes, a quite different route from this section's. Check its answer numerically.

3. **Another quartic.** Use the method of this section to solve $x^4 - 4x^3 + 7x^2 - 6x + 2 = 0$. (One root of the resolvent cubic is a small integer.)

4. **Counting permutations.** Of the 24 permutations of $\alpha, \beta, \gamma, \delta$, find all those that send the pairing $\{\alpha\beta, \gamma\delta\}$ to itself. Show that there are 8 of them. (They form a subgroup of S_4.)

Further reading

- **Mario Livio, *The Equation That Couldn't Be Solved*** (Simon & Schuster). The story of the quintic and the birth of group theory, for a general reader.
- **Ian Stewart, *Why Beauty Is Truth: A History of Symmetry*** (Basic Books). From Babylonian equations to Galois and beyond.

- **Ian Stewart, *Galois Theory*** (CRC Press). The undergraduate text, with a historical introduction covering exactly this material.

The full guided solution is in Section 6.4:
https://beyondstep.keithmcnulty.org/ch06/quartic.html